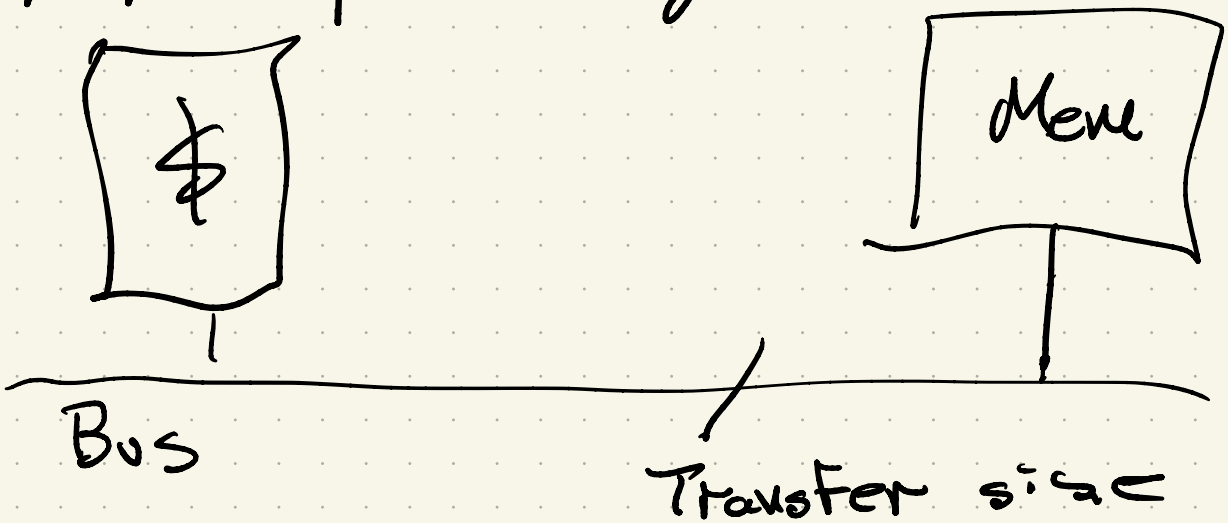
Miss penalty



$$\frac{2^B}{\text{Transfer size}} \quad \text{Bus transactions}$$

$$\text{Transaction time} \times \left(\frac{2^B}{\text{Transfer size}}\right)$$

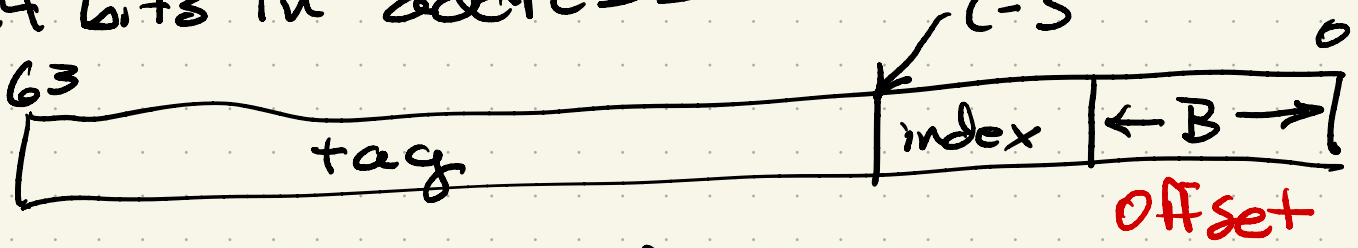
Pipelined bus: Transfer full block in a stream

$$\text{Transaction time} + \left(\frac{2^B}{\text{Transfer size}} - 1\right) \text{subsequent time}$$

$$10\,ns + \left(\frac{64}{16} - 1\right) \times 1\,ns = 13\,ns$$

Tag store size related to block size

64 bits in address



$64-(C-S)$

All blocks have status info

WB/WA bit for dirty

Always have a "valid" bit

Tag is $64-(C-S)+1\ (+1)$ (valid; Dirty (WB/WA))

1 tag per block, $\frac{2^C}{2^B}$ total blocks

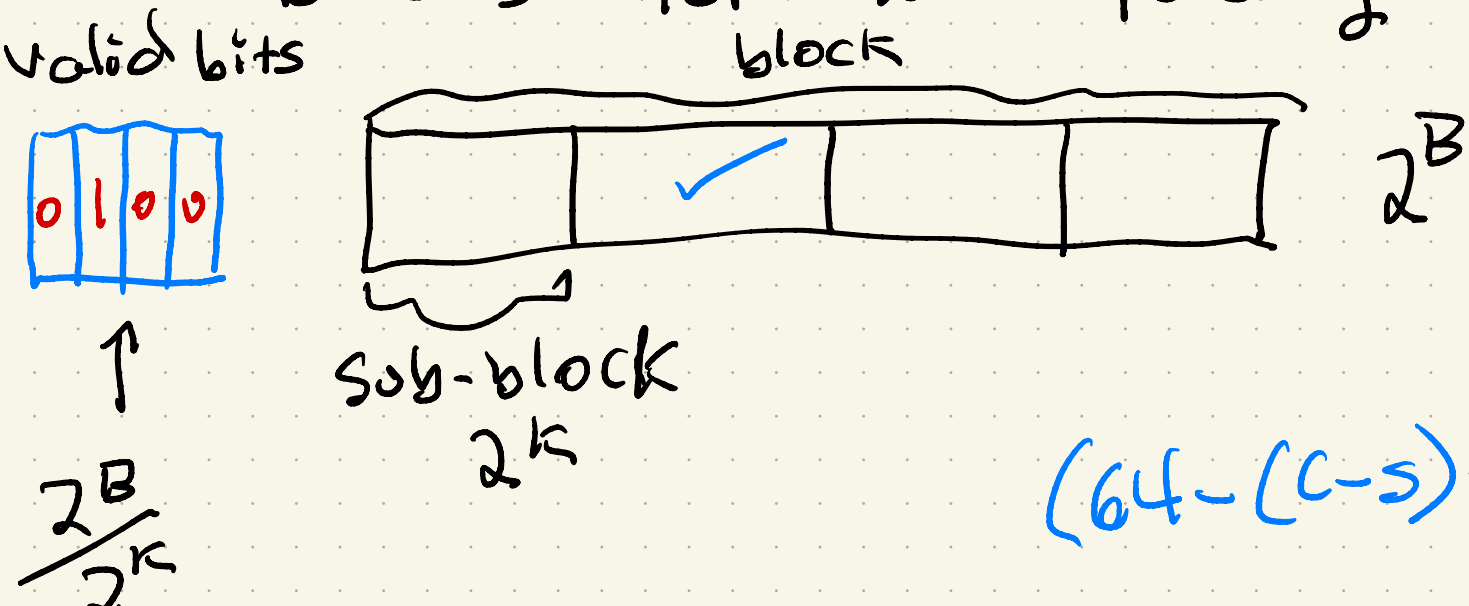
Tag store size in bits $= (64-(C-S)+1)\cdot 2^{C-B}$

$\Rightarrow$ large blocks mean less tag storage

Problem with large blocks is larger miss penalty

$$AAT = HT + MR \times MP$$

In summary: want large blocks for locality, small tag storage, but large blocks hurt miss penalty



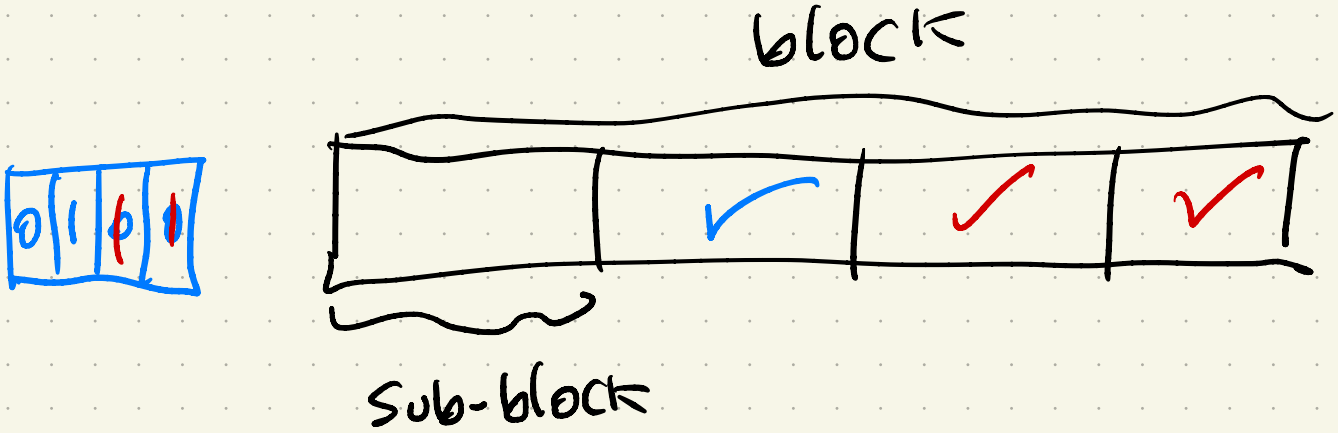
$$(64-(C-S)+1)\,2^{C-K}$$

$$\text{Tag store size} = \left(64-(C-S)+\frac{2^B}{2^K}\right) \times 2^{C-B}$$

Miss penalty only depends on sub-block

Adding back locality:

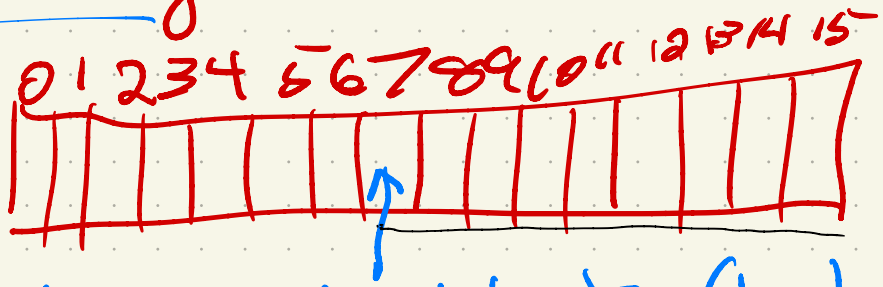
Spatial locality has a special case "sequential locality"



Fetch sub-block, give to CPU
then fetch rest of sub-blocks in block

Even without sub-blocking, can use this:

Old way: Plain old cache



$2^4 = 2^B$

Fetch whole block (bytes 0 thru 15)
Put in $, then give CPU byte 7

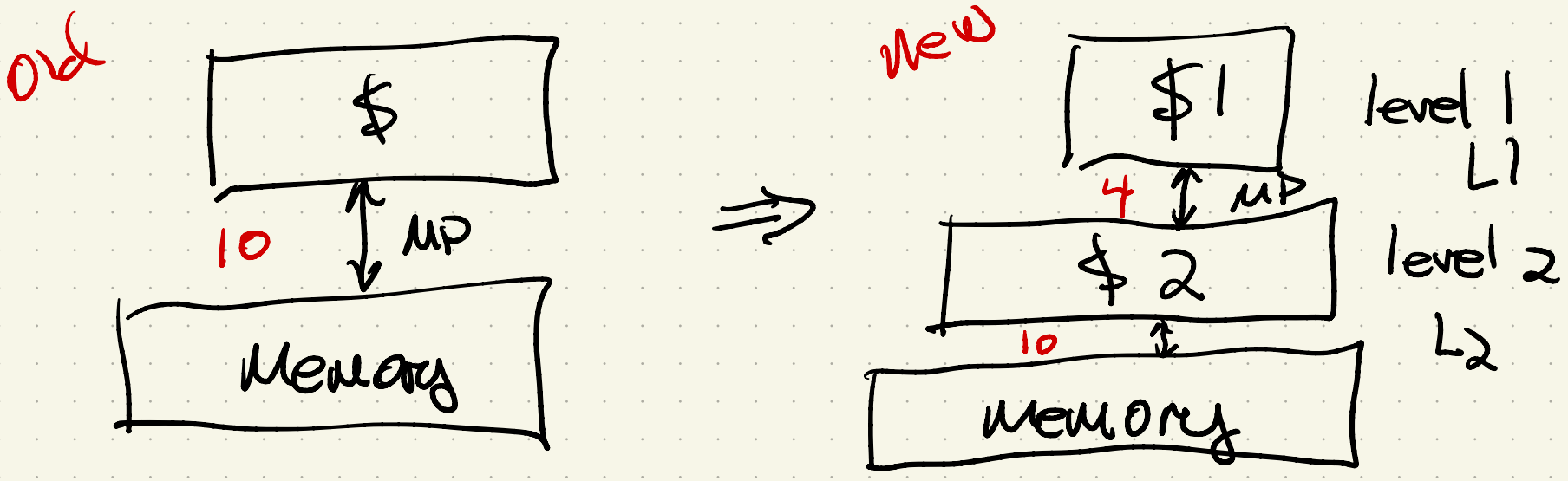
1) Instead fetch starting with byte 7

0, 1, 2 ··· 14, 15 → 7, 8, 9 ··· 14, 15, 0,
Critical word first 1, 2 ··· 6

2) Give CPU byte 7 as soon as it arrives

Early restart

Miss penalty reduction: L2 cache



$$AAT = HT_1 + MR_1 \times MP_1$$

$$MP_1 = HT_2 + MR_2 \times MP_2$$

$MR_1 = 5\%$, $HT_1 = 2$

$AAT = 2 + 0.05 \times 10$

$= 2.5$

old

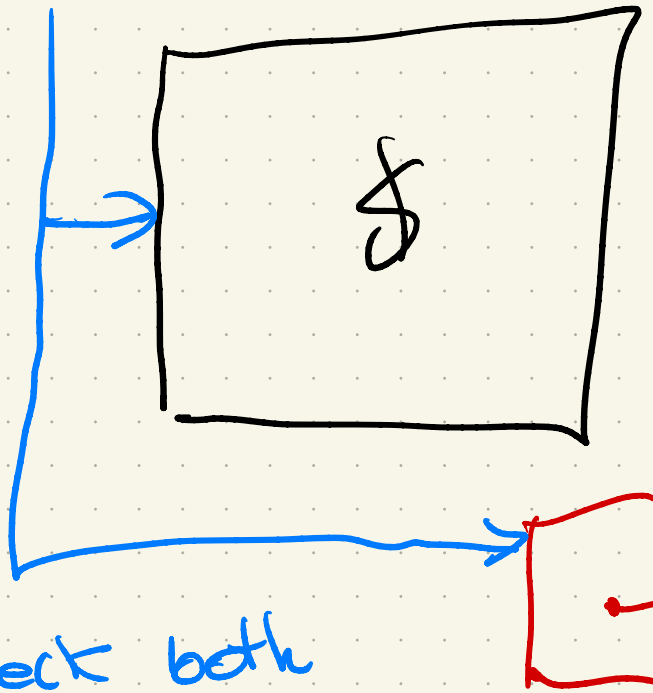
$HT_2 = 4$ $MR_2 = 1\%$

$AAT_2 = MP_1 = 4 + 0.01 \times 10$

$= 4.1$

$AAT = 2 + 0.05 \times (4.1)$

$= 2.21$



Steps in a miss:

1. Pick a block to evict "victim"
2. Get missing block
3. ...

Ex: $MR_{VC} = 20\%$

$AAT = 2 + 0.05(0.2 \cdot 10) = 2.1$

if hit in victim cache, give to CPU,
put victim block in main $
put new victim in victim cache

$$AAT = HT + MR_1 \times (MR_{VC} \times MP)$$

$$MR_1 = \frac{\#\text{ misses}_1}{\text{total accesses}}, \quad MR_{VC} = \frac{\#\text{ VC misses}}{\#\text{ misses}_1}$$